Technical report: EEG electrode impedance under VR headsets

BrainAccess MINI (8 ch), MIDI (16 ch) and MAXI (32 ch) with HTC Vive and Meta Quest

Date: 2026-11-06
Prepared by: EEG-VR-stats project
Data: 1 participant, session S001; 3 recordings per device (no VR, HTC, Meta Quest)

1 Summary

- **VR headsets did not raise impedance overall.** Median impedance across electrodes (no VR / HTC Vive / Meta Quest) was 87 / 83 / 83 kΩ (MINI), 84 / 124 / 61 kΩ (MIDI) and 135 / 90 / 98 kΩ (MAXI). Only MAXI shows a systematic difference (lower with VR), but its no-VR recording was the first after fitting and impedance was still falling steeply (−31.6 kΩ/min), so this most likely reflects settling rather than the headsets.
- **Midline electrodes were much lower under both VR headsets** on MIDI and MAXI (Fz 255 → 19 / 19 kΩ on MIDI, 264 → 33 / 37 kΩ on MAXI; HTC / Meta Quest). Other electrodes changed in both directions.
- **Electrodes to check:** MIDI C4 was above 1000 kΩ in every recording (1272, 1218, 1287 kΩ). On MAXI, 6 electrodes had a median above 500 kΩ without VR (FC2, P3, POz, P4, F7, FC6).
- **Settling.** Impedance fell during the first minutes with MINI and especially MAXI; with MAXI only 6 of 32 electrodes were stable from 60 s onwards in the first (no-VR) recording. MIDI was stable from early on.
- **Short spikes** occurred with MINI (up to 8288 kΩ) and in one MAXI recording (1347 kΩ). **A 5-s rolling median** removes them and keeps real changes.

2 Setup

	MINI	**MIDI**	**MAXI**
Device	BrainAccess MINI	BrainAccess MIDI	BrainAccess MAXI
Electrodes (n)	8: F3, F4, C3, C4, P3, P4, O1, O2	16: F7, F3, Fz, F4, F8, C3, Cz, C4, P7, P3, Pz, P4, P8, O1, Oz, O2	32: AF3, AFz, AF4, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, C3, Cz, C4, T7, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO3, POz, PO4, O1, Oz, O2
Sampling rate	250 Hz	250 Hz	250 Hz
Recording order	No VR, HTC Vive, Meta Quest	No VR, Meta Quest, HTC Vive	No VR, HTC Vive, Meta Quest
Start (min)*	0 / 6 / 12	0 / 33 / 27	0 / 5 / 10
Duration (s)*	154 / 156 / 168	149 / 155 / 154	141 / 150 / 157

*no VR / HTC Vive / Meta Quest; start relative to the first recording with that device.

All recordings were made in impedance mode with dry electrodes and one participant (sub-1) on the same day; each device was fitted separately, and the VR headset was worn over the EEG headset. Recordings were exported from the acquisition software (SQLite) to CSV; dates, device serial numbers and computer timestamps were removed. No data were excluded or interpolated.

3 Processing

- Each electrode's impedance was smoothed with a centred rolling median of 1, 5, 15, 30 s. The median ignores isolated spikes.
- **Initial / final impedance:** median of the first / last 10 s.
- **Settling time:** last moment the 5-s median was more than max(10%, 10 kΩ) away from the final value.
- **Trend:** robust (Theil–Sen) slope of the impedance over time, in kΩ/min.
- **High impedance:** electrode median above 500 kΩ (a reporting threshold, not a device specification).

4 Results

4.1 Device overview

On the 8 electrodes present on all devices (F3, F4, C3, C4, P3, P4, O1, O2), median impedance without VR was similar for MINI and MIDI (87 and 84 kΩ) and higher for MAXI (260 kΩ); see Table 1. Each device was fitted separately, so this compares fittings as much as devices.

Table 1: Median over electrodes of the per-electrode median impedance (kΩ): on the 8 common electrodes and on all electrodes of each device (number of electrodes in brackets).

	No VR		HTC Vive		Meta Quest	
Device	common (8)	all	common (8)	all	common (8)	all
BA_MAXI_32ch	260	135 (32)	136	90 (32)	186	98 (32)
BA_MIDI_16ch	84	84 (16)	161	124 (16)	113	61 (16)
BA_MINI_8ch	87	87 (8)	83	83 (8)	83	83 (8)

4.2 MINI (8 channels)

- Electrode medians ranged from 51 (P4) to 152 kΩ (F4) without VR, 14 (O2) to 226 kΩ (P3) with HTC and 27 (O1) to 225 kΩ (C3) with Meta Quest; no consistent shift between conditions (Friedman $p = 0.325$).
- Impedance fell by more than 10% in 5, 5 and 5 of 8 electrodes (no VR, HTC, Meta Quest), mostly in steps. Largest drops: F4 $1119 \to 149$, P3 $378 \to 92$, C3 $337 \to 96$ kΩ.
- Median settling time 119, 35 and 119 s; only 2, 5 and 2 electrodes were stable from 60 s onwards.
- Spikes up to 8288 kΩ (F3, HTC) and 4357 kΩ (F4, Meta Quest).

4.3 MIDI (16 channels)

- Electrode medians ranged from 50 (P7) to 1272 kΩ (C4) without VR, 19 (Fz) to 1218 kΩ (C4) with HTC and 16 (O1) to 1287 kΩ (C4) with Meta Quest; no consistent shift between conditions (Friedman $p = 0.133$).
- **Midline electrodes** were much lower with both VR headsets than without: Fz 255 vs. 19 (HTC) and 19 kΩ (Meta Quest), Cz 75 vs. 27 and 23, Pz 69 vs. 23 and 28 kΩ.

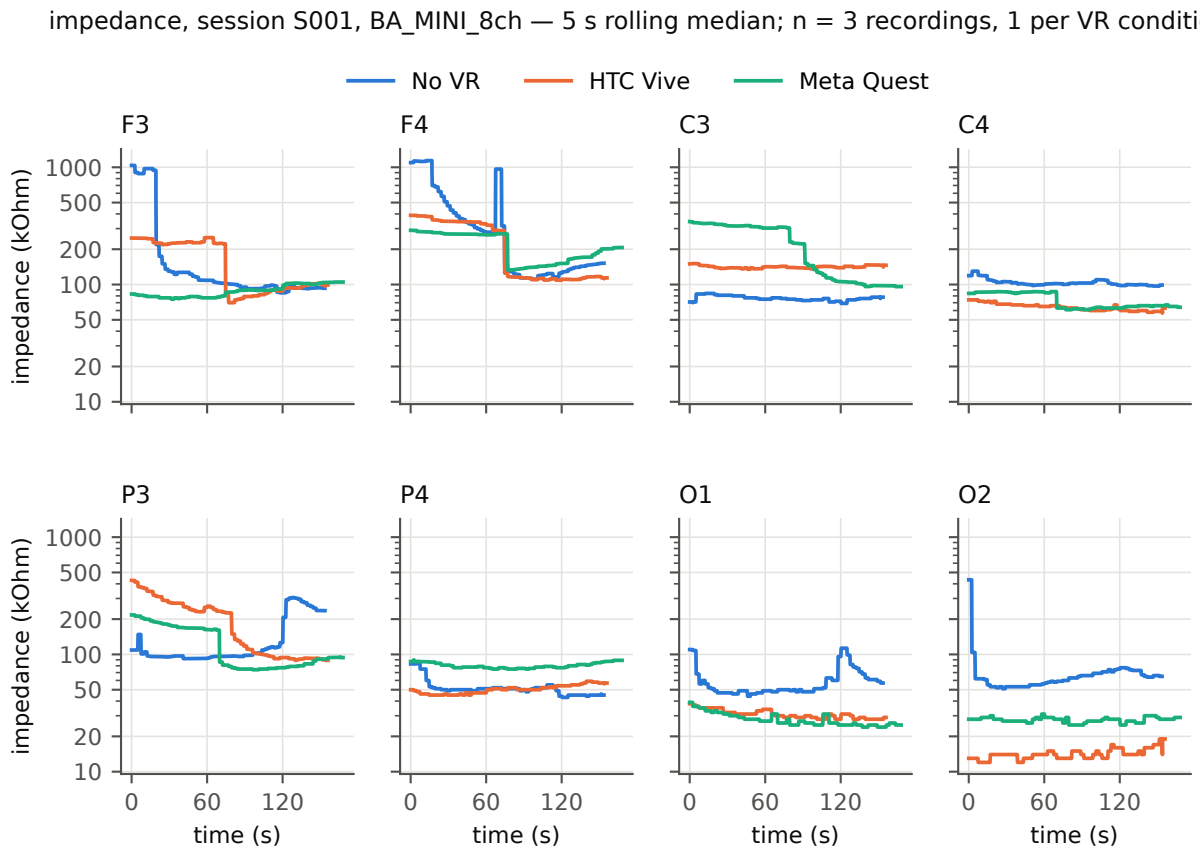


Figure 1: MINI: impedance (kΩ, log scale) per electrode, 5-s rolling median, one recording per condition (250 Hz; $n = 38406, 39000, 42000$ samples).

Table 2: MINI: impedance per electrode (kΩ): median [interquartile range], initial → final value, and settling time (s).

	No VR			HTC Vive			Meta Quest		
Ch.	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)
F3	103 [94–127]	902 → 93	70	102 [87–227]	248 → 101	120	87 [79–100]	81 → 105	120
F4	152 [128–431]	1119 → 149	127	119 [113–344]	385 → 115	75	206 [151–270]	284 → 206	149
C3	76 [74–79]	81 → 78	0	141 [139–143]	150 → 146	0	225 [105–316]	337 → 96	118
C4	102 [100–106]	120 → 98	111	64 [61–67]	74 → 59	22	66 [63–86]	85 → 65	70
P3	98 [95–126]	109 → 237	142	226 [94–267]	378 → 92	99	92 [78–170]	212 → 94	144
P4	51 [48–52]	83 → 45	15	50 [47–54]	48 → 57	48	79 [77–84]	87 → 89	125
O1	51 [48–61]	68 → 60	137	31 [29–33]	36 → 28	0	27 [25–30]	36 → 25	7
O2	63 [55–69]	64 → 65	127	14 [13–15]	13 → 17	0	28 [26–29]	28 → 28	0

- **Lateral fronto-central electrodes** were higher with both headsets: F3 58 vs. 151 / 145, F4 71 vs. 212 / 150, C3 85 vs. 185 / 121 kΩ (HTC / Meta Quest).
- **Posterior electrodes** were lowest with Meta Quest (P7 18, P8 17, O1 16, O2 19 kΩ), while P7 was highest with HTC (192 kΩ).
- **C4** stayed at 1272, 1218 and 1287 kΩ in all three recordings. On MAXI, C4 was normal (115–220 kΩ), which points to the MIDI electrode or its connector rather than the position.
- Impedance was stable: median settling time 57, 65 and 0 s; 8, 6 and 11 of 16 electrodes were stable from 60 s onwards. No spikes comparable to MINI occurred.

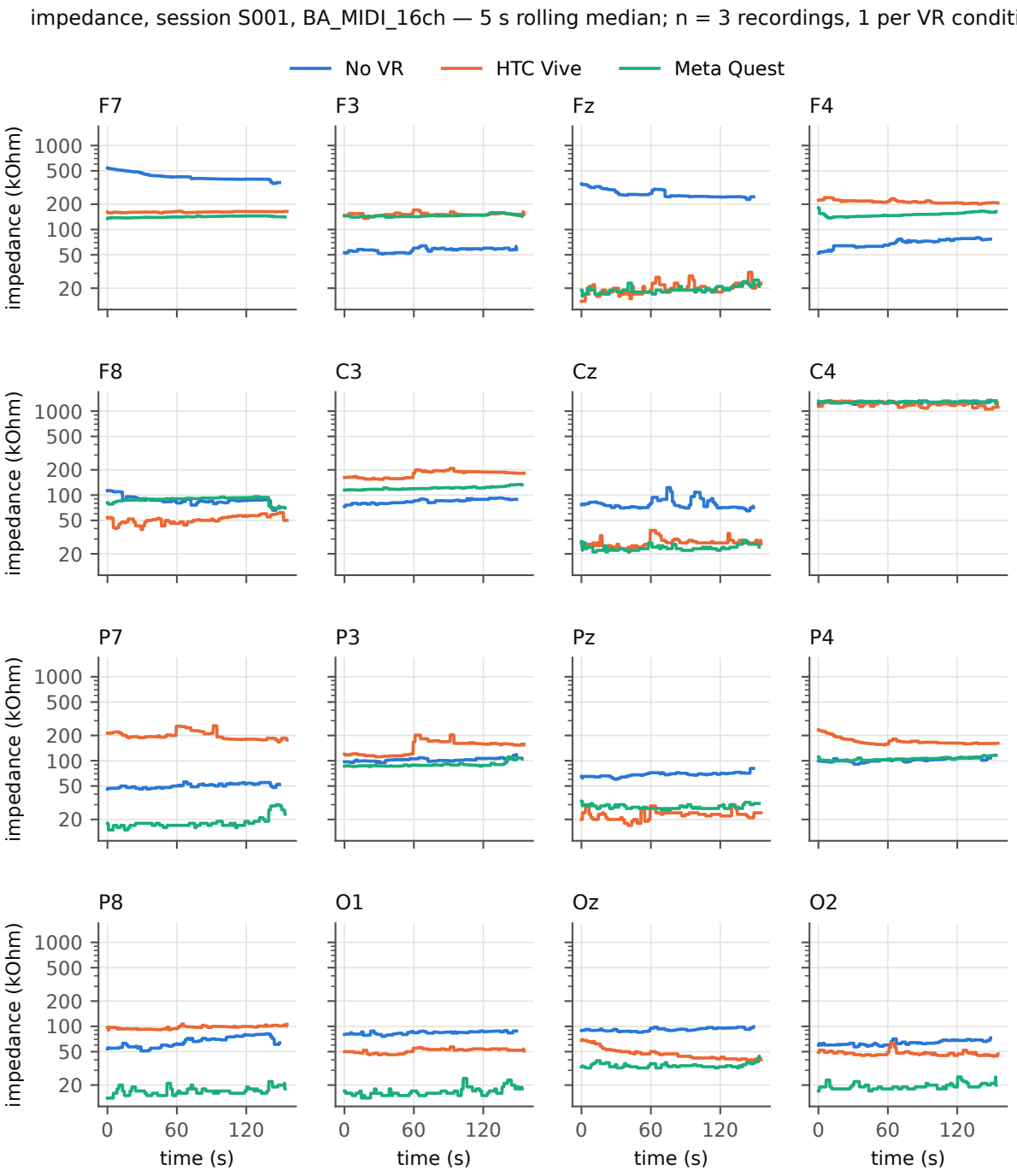


Figure 2: MIDI: impedance (kΩ, log scale) per electrode, 5-s rolling median, one recording per condition (250 Hz; $n = 37200, 38800, 38400$ samples).

Table 3: MIDI: impedance per electrode (kΩ): median [interquartile range], initial → final value, and settling time (s).

	No VR			HTC Vive			Meta Quest		
Ch.	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)
F7	406 [398–440]	524 → 362	107	162 [160–163]	161 → 163	0	142 [139–144]	138 → 142	0
F3	58 [55–60]	55 → 60	0	151 [149–156]	154 → 152	64	145 [143–148]	143 → 150	0
Fz	255 [246–296]	332 → 243	72	19 [18–22]	17 → 22	0	19 [18–20]	19 → 22	0
F4	71 [63–76]	55 → 76	61	212 [207–219]	238 → 208	66	150 [145–155]	148 → 161	46
F8	86 [83–89]	111 → 71	141	52 [48–56]	45 → 60	96	91 [87–93]	82 → 70	139
C3	85 [81–89]	77 → 89	29	185 [161–190]	164 → 183	94	121 [117–124]	116 → 133	72
Cz	75 [71–82]	79 → 70	115	27 [25–29]	26 → 27	64	23 [22–25]	24 → 26	0
C4	1272 [1230–1312]	1253 → 1278	0	1218 [1142–1279]	1261 → 1059	155	1287 [1247–1322]	1312 → 1330	0
P7	50 [48–53]	47 → 52	0	192 [183–213]	217 → 176	94	18 [17–19]	16 → 29	126
P3	102 [99–106]	97 → 105	149	158 [118–164]	120 → 155	94	88 [87–91]	86 → 107	139
Pz	69 [66–71]	65 → 72	32	23 [21–24]	23 → 24	0	28 [27–30]	30 → 31	0
P4	102 [99–106]	98 → 107	77	164 [161–172]	223 → 161	67	105 [102–108]	102 → 115	69
P8	65 [57–74]	55 → 71	139	98 [94–100]	97 → 103	54	17 [16–18]	16 → 19	0
O1	84 [81–87]	81 → 88	35	52 [49–54]	50 → 52	0	16 [15–18]	16 → 19	0
Oz	91 [89–95]	90 → 96	53	47 [42–51]	66 → 40	34	34 [32–35]	33 → 37	0
O2	63 [60–67]	60 → 68	34	47 [46–49]	50 → 45	67	19 [18–21]	19 → 21	0

4.4 MAXI (32 channels)

- Electrode medians ranged from 34 (AF4) to 1434 kΩ (FC2) without VR, 13 (AFz) to 595 kΩ (PO3) with HTC and 19 (AFz) to 670 kΩ (FC6) with Meta Quest. Impedance was lower in both VR recordings than in the no-VR recording (Friedman $p = 0.003$).
- **This difference is confounded with settling.** The no-VR recording came first and impedance was falling in 21 of 32 electrodes (median trend −31.6 kΩ/min; median settling time 93 s; only 6 electrodes stable from 60 s). In the later VR recordings trends were small (−4.4 and −3.7 kΩ/min) and 18 and 21 electrodes were stable from 60 s.
- **Midline and occipital electrodes** were much lower with both VR headsets: Fz 264 vs. 33 / 37, Cz 138 vs. 34 / 43, O1 132 vs. 30 / 32, O2 283 vs. 48 / 45 kΩ (HTC / Meta Quest).
- **High-impedance electrodes** (median above 500 kΩ): 6 without VR (FC2, P3, POz, P4, F7, FC6), 1 with HTC (PO3) and 1 with Meta Quest (FC6). CP2 showed repeated short jumps without VR (Fig. 3), typical of intermittent contact.
- One spike up to 1347 kΩ (FC2, HTC); the highest 1-s median in that recording was 965 kΩ.

4.5 Choice of smoothing window

Rolling medians of all four lengths agree for most of the time and differ only around spikes and short changes (Fig. 4 and Appendix A). Tables 5–7 and Fig. 5 summarise the effect:

- For stable recordings the trend estimate barely depends on the window (1-s vs. 30-s: −3.3 vs. −3.3 for MINI and +3.2 vs. +3.0 kΩ/min for MIDI, no VR). For the steeply falling MAXI no-VR recording it changed from −30.4 to −48.2 kΩ/min, the 30-s value resting on only 4 points; use windows of ≤15 s for trends over recordings of this length.
- With MINI, a 1-s window removed the HTC spike (8288 kΩ) but not the Meta Quest excursion (4357 kΩ); a 5-s window removed both.
- 15–30 s windows also remove genuine short changes, e.g. the brief rise of F4 at about 65 s in the MINI no-VR recording, visible only at 1 and 5 s.

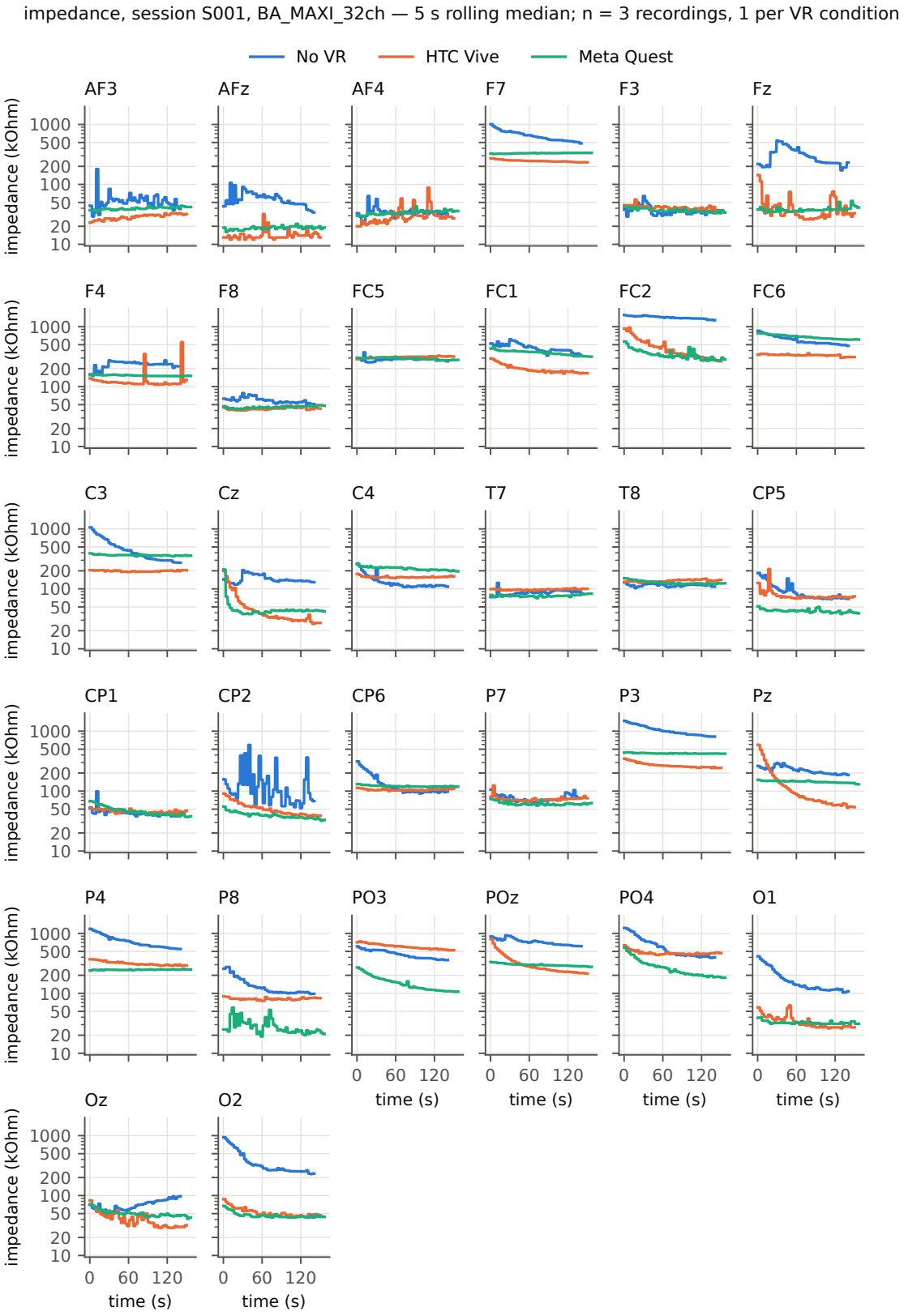


Figure 3: MAXI: impedance (kΩ, log scale) per electrode, 5-s rolling median, one recording per condition (250 Hz; $n = 35200, 37600, 39200$ samples).

Table 4: MAXI: impedance per electrode (kΩ): median [interquartile range], initial → final value, and settling time (s).

	No VR			HTC Vive			Meta Quest		
Ch.	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)	median [IQR]	init. → final	settle (s)
AF3	51 [45–57]	34 → 43	134	29 [26–31]	24 → 32	0	40 [38–41]	37 → 42	0
AFz	56 [47–66]	47 → 36	125	13 [13–15]	13 → 15	64	19 [18–19]	17 → 19	0
AF4	34 [32–36]	29 → 32	32	27 [24–30]	20 → 29	115	34 [32–35]	30 → 36	0
F7	629 [551–745]	930 → 506	93	244 [240–252]	270 → 233	19	331 [328–333]	321 → 335	0
F3	36 [33–40]	38 → 36	38	41 [39–43]	45 → 36	19	37 [35–39]	42 → 34	0
Fz	264 [226–354]	208 → 191	141	33 [29–44]	112 → 33	138	37 [37–39]	38 → 43	0
F4	237 [219–254]	160 → 219	118	114 [111–119]	131 → 129	144	153 [151–156]	159 → 151	0
F8	59 [55–62]	61 → 51	61	43 [42–44]	45 → 44	0	45 [44–47]	44 → 49	0
FC5	287 [276–291]	291 → 275	13	314 [307–317]	293 → 320	3	294 [288–303]	297 → 279	61
FC1	447 [404–497]	497 → 351	125	182 [174–206]	284 → 168	83	375 [341–390]	432 → 317	109
FC2	1434 [1374–1494]	1540 → 1306	77	372 [314–558]	927 → 267	134	322 [299–378]	519 → 283	138
FC6	552 [530–670]	821 → 494	83	336 [332–340]	350 → 311	80	670 [627–722]	778 → 610	80
C3	391 [304–554]	972 → 275	109	198 [195–203]	205 → 204	0	364 [359–367]	381 → 358	0
Cz	138 [134–156]	130 → 130	74	34 [30–48]	161 → 27	61	43 [41–44]	74 → 43	10
C4	115 [112–132]	236 → 112	51	157 [155–159]	171 → 161	0	220 [208–228]	241 → 199	86
T7	86 [81–92]	74 → 89	29	98 [96–99]	99 → 100	0	76 [75–78]	73 → 83	83
T8	115 [110–119]	119 → 109	61	135 [131–140]	132 → 139	0	124 [122–132]	149 → 123	29
CP5	77 [70–108]	167 → 70	61	73 [72–76]	94 → 73	26	43 [41–44]	47 → 40	3
CP1	43 [41–45]	43 → 39	13	45 [44–48]	50 → 45	0	44 [41–50]	67 → 37	51
CP2	96 [69–155]	130 → 73	134	47 [41–56]	84 → 39	74	39 [36–42]	48 → 33	26
CP6	104 [97–140]	273 → 98	118	105 [103–108]	111 → 108	0	120 [119–122]	130 → 119	3
P7	72 [69–84]	91 → 77	131	73 [70–75]	82 → 77	6	60 [59–62]	72 → 62	3
P3	989 [879–1193]	1427 → 813	96	261 [252–282]	336 → 244	58	424 [421–430]	443 → 421	0
Pz	221 [198–247]	247 → 193	80	74 [61–131]	490 → 55	106	145 [140–147]	151 → 131	83
P4	683 [602–823]	1131 → 551	93	307 [298–331]	368 → 292	51	248 [246–249]	247 → 250	0
P8	109 [103–152]	274 → 98	70	81 [79–83]	88 → 83	0	25 [23–30]	25 → 22	77
PO3	442 [386–525]	574 → 359	90	595 [556–654]	714 → 523	93	136 [115–173]	257 → 107	106
POz	723 [653–788]	850 → 614	96	260 [230–343]	678 → 217	102	297 [290–304]	329 → 277	80
PO4	466 [426–730]	1202 → 396	99	460 [453–477]	584 → 477	13	231 [200–300]	513 → 182	122
O1	132 [116–200]	373 → 104	131	30 [27–39]	51 → 27	80	32 [31–33]	39 → 31	0
Oz	70 [58–82]	62 → 95	118	36 [31–49]	69 → 30	90	49 [46–52]	68 → 43	35
O2	283 [259–401]	870 → 233	131	48 [47–56]	78 → 47	35	45 [44–47]	63 → 44	16

Table 5: MINI, effect of window length: average removed variation (kΩ) and trend (kΩ/min), median over 8 electrodes.

	No VR		HTC Vive		Meta Quest	
Window (s)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)
1	0.1	−3.3	0.0	−4.1	0.0	−5.8
5	1.8	−3.5	1.4	−4.1	1.2	−5.9
15	3.3	−3.5	2.0	−4.4	1.3	−6.6
30	5.8	−3.3	2.2	−4.5	1.6	−7.9

Table 6: MIDI, effect of window length: average removed variation (kΩ) and trend (kΩ/min), median over 16 electrodes.

	No VR		HTC Vive		Meta Quest	
Window (s)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)
1	0.0	+3.2	0.0	+1.2	0.0	+2.1
5	1.2	+3.2	1.6	+1.2	1.0	+1.8
15	1.9	+3.4	2.4	+0.8	1.4	+1.7
30	2.1	+3.0	2.8	+1.2	1.5	+2.0

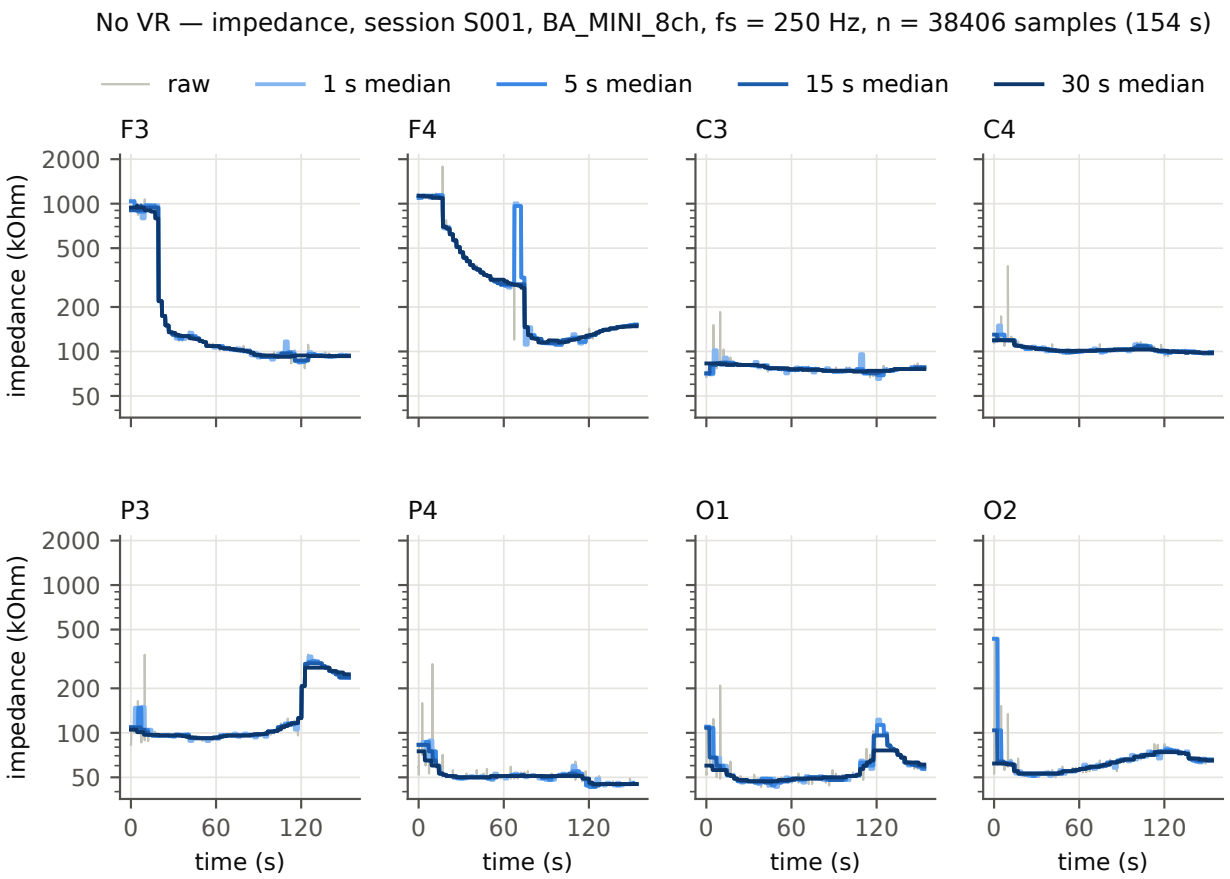
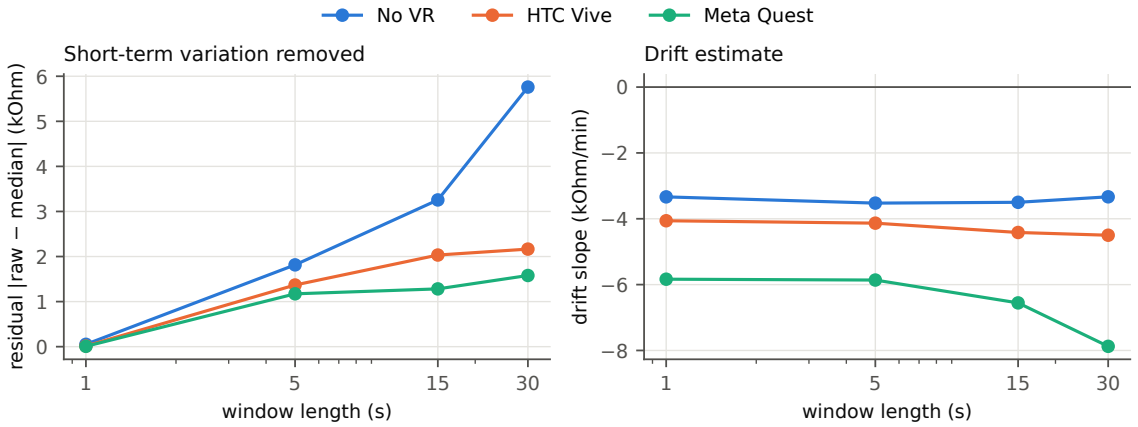


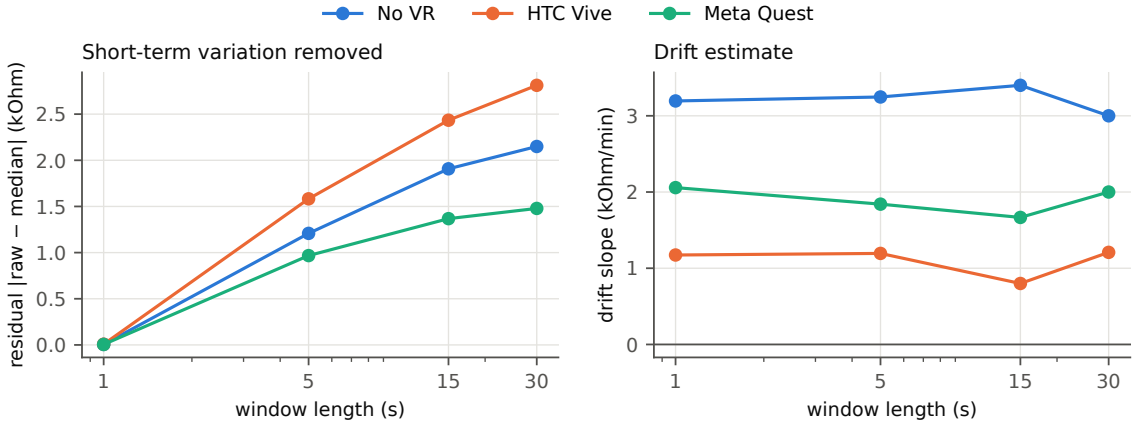
Figure 4: MINI, no-VR recording: raw impedance (grey) and rolling medians of 1, 5, 15, 30 s (light to dark blue). Log scale.

Table 7: MAXI, effect of window length: average removed variation (kΩ) and trend (kΩ/min), median over 32 electrodes.

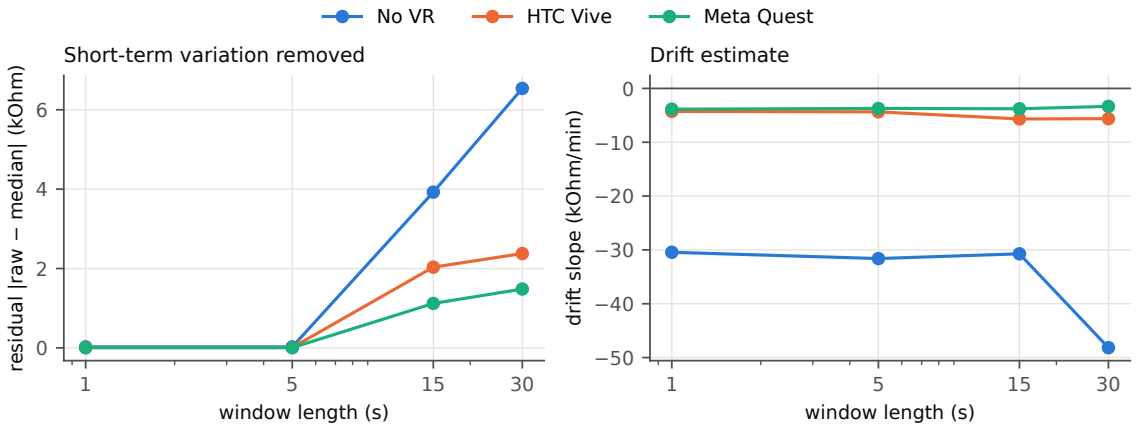
	No VR		HTC Vive		Meta Quest	
Window (s)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)	resid. (kΩ)	slope (kΩ/min)
1	0.0	−30.4	0.0	−4.3	0.0	−3.9
5	0.0	−31.6	0.0	−4.4	0.0	−3.7
15	3.9	−30.7	2.0	−5.7	1.1	−3.8
30	6.5	−48.2	2.4	−5.6	1.5	−3.3



Median over n = 8 channels per recording; impedance, session S001, BA_MINI_8ch.



Median over n = 16 channels per recording; impedance, session S001, BA_MIDI_16ch.



Median over n = 32 channels per recording; impedance, session S001, BA_MAXI_32ch.

Figure 5: Effect of window length, MINI (top), MIDI (middle) and MAXI (bottom). Left: average difference between raw and smoothed impedance (what the window removes). Right: trend estimate. Median over electrodes per recording.

5 Recommendations

1. Display or log impedance with a **5-s rolling median**; use windows up to 15 s for reporting trends.
2. **Check hardware/fit:** MIDI C4 (above 1000 kΩ in every recording, normal on MAXI); on MAXI the electrodes above 500 kΩ (FC2, P3, POz, P4, F7, FC6) and CP2 (intermittent contact).
3. Allow a **settling period** after fitting the EEG headset and re-check impedance before recording. With MINI and MAXI, 2.5–3 minutes were not enough for all electrodes.
4. Re-check impedance **after putting on the VR headset**: it changed individual electrodes in both directions (midline lower on MIDI and MAXI; some lateral electrodes higher), so a pre-VR check is not sufficient.
5. For a fair VR comparison, **record a settled no-VR baseline** (or randomise the order of conditions); in this data set the no-VR recording always came first.

6 Limitations

One participant, one recording per condition and device. Conditions were recorded in a fixed order that differed between devices (MINI: No VR, HTC Vive, Meta Quest; MIDI: No VR, Meta Quest, HTC Vive; MAXI: No VR, HTC Vive, Meta Quest), always starting without VR, so differences between conditions may also reflect time since fitting or re-fitting of the headsets. Results should be confirmed with more participants and a balanced order of conditions.

A Window comparison, remaining recordings

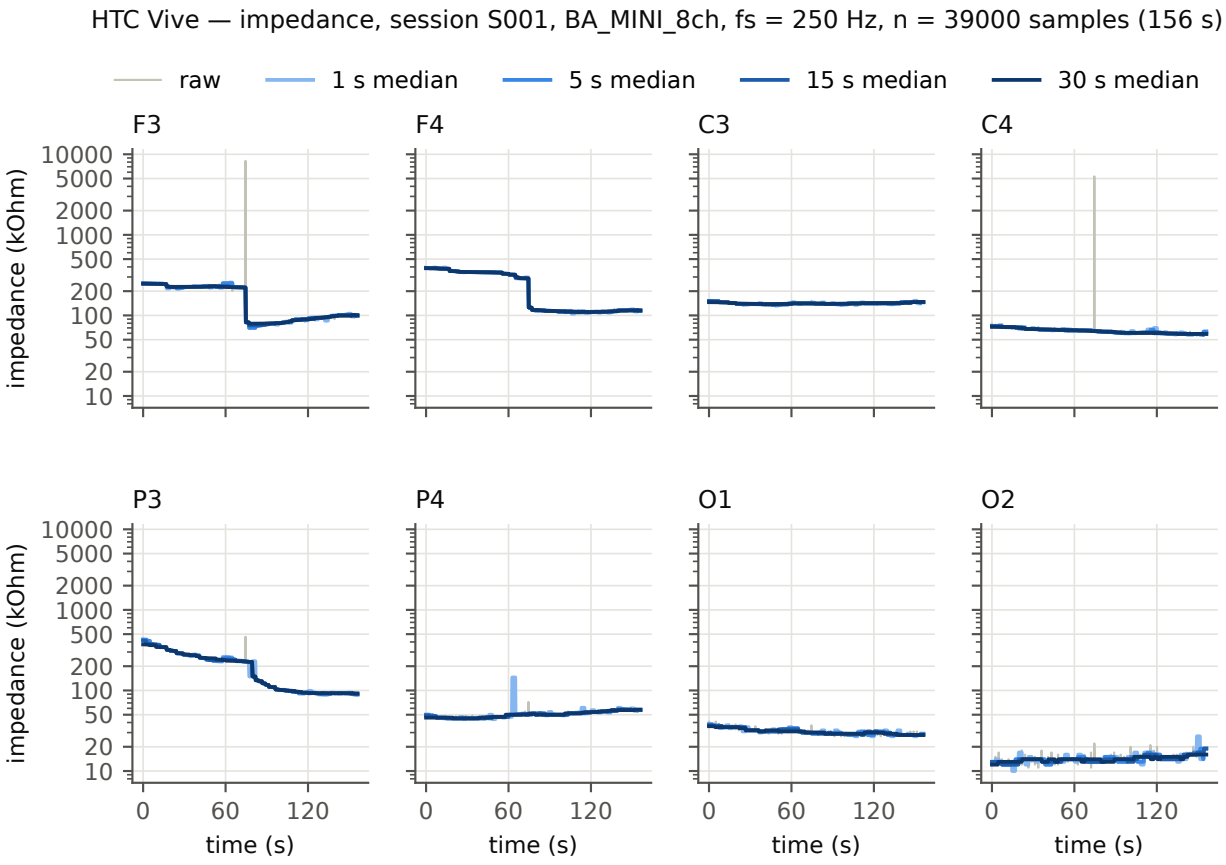


Figure 6: MINI, HTC Vive; as Fig. 4.

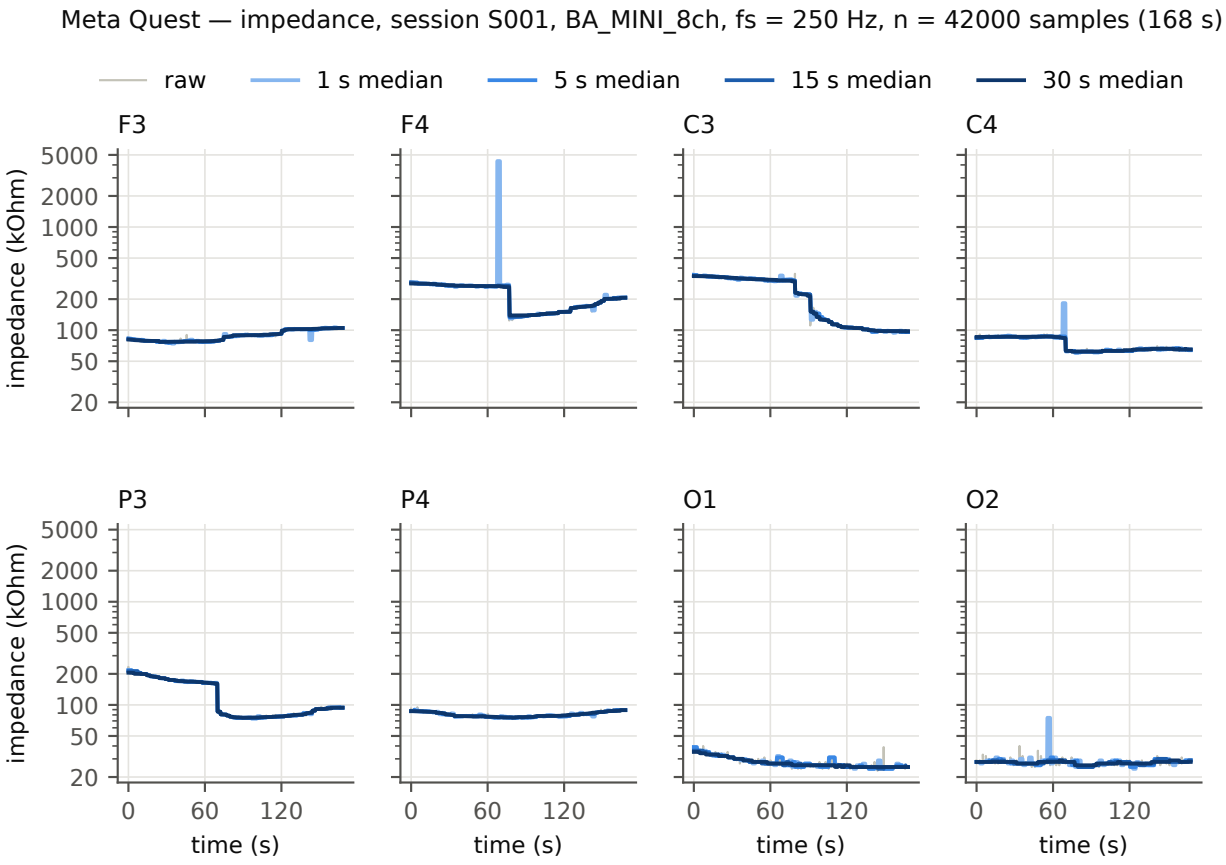


Figure 7: MINI, Meta Quest; as Fig. 4.

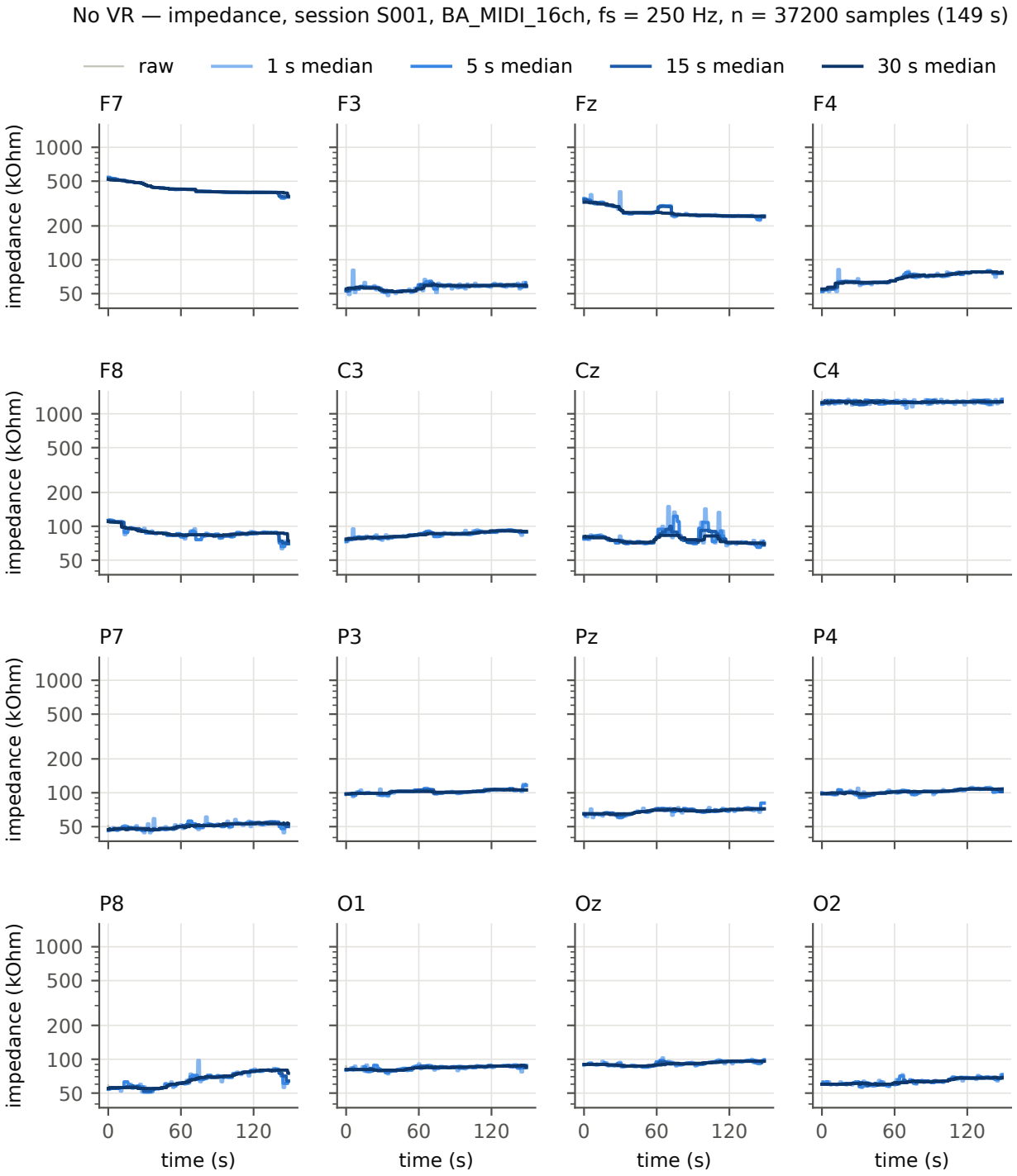


Figure 8: MIDI, no VR; as Fig. 4.

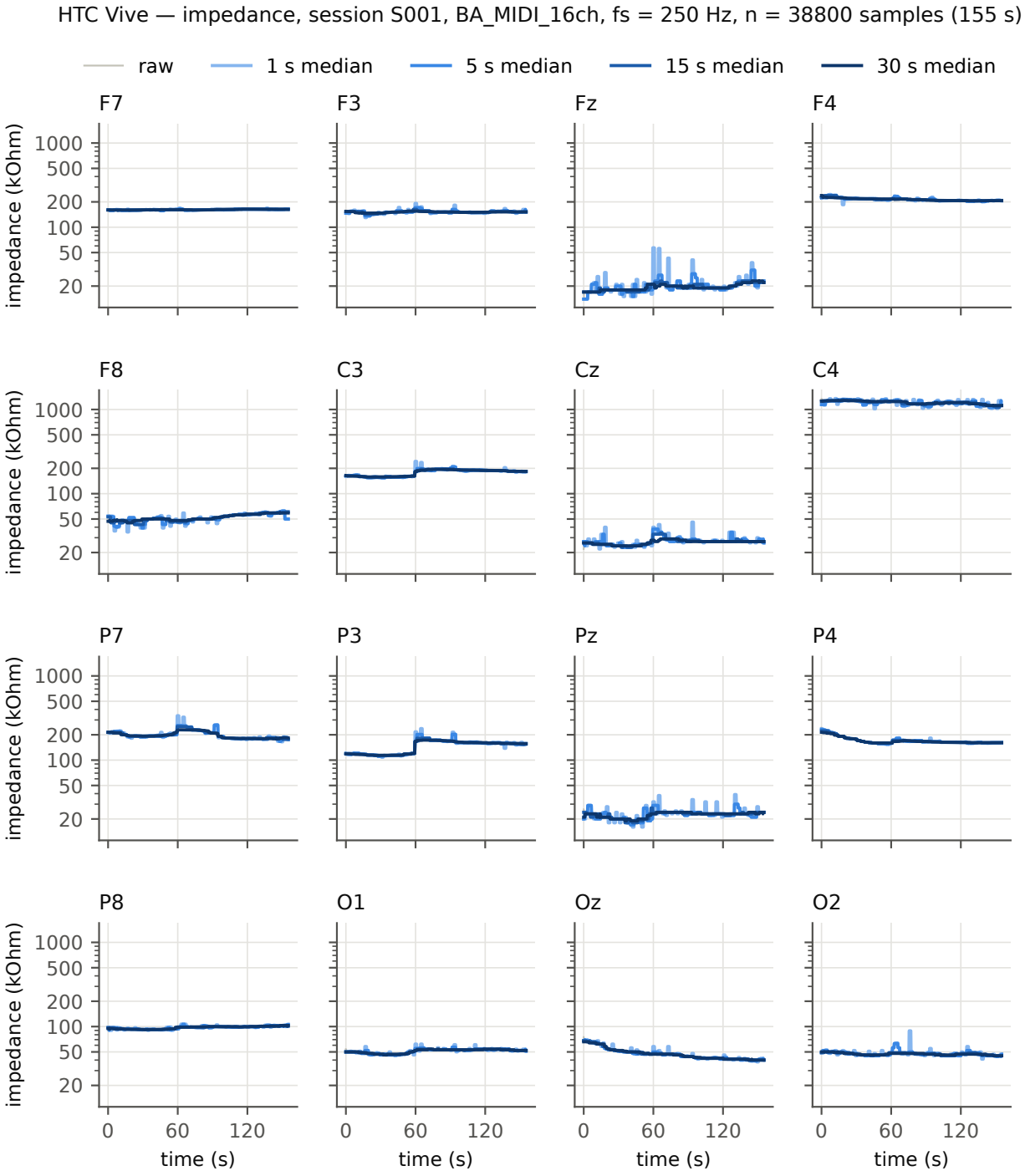


Figure 9: MIDI, HTC Vive; as Fig. 4.

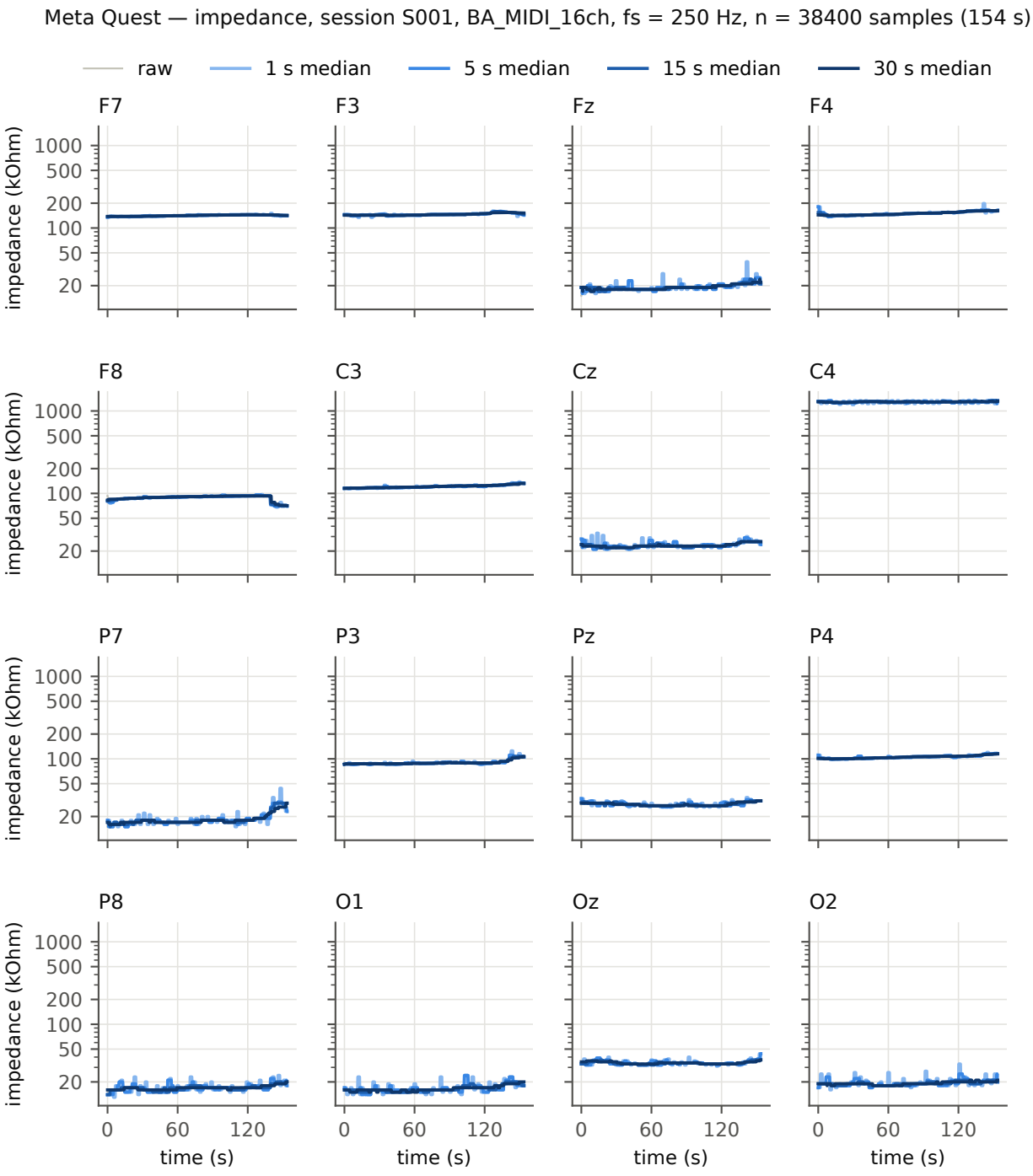


Figure 10: MIDI, Meta Quest; as Fig. 4.

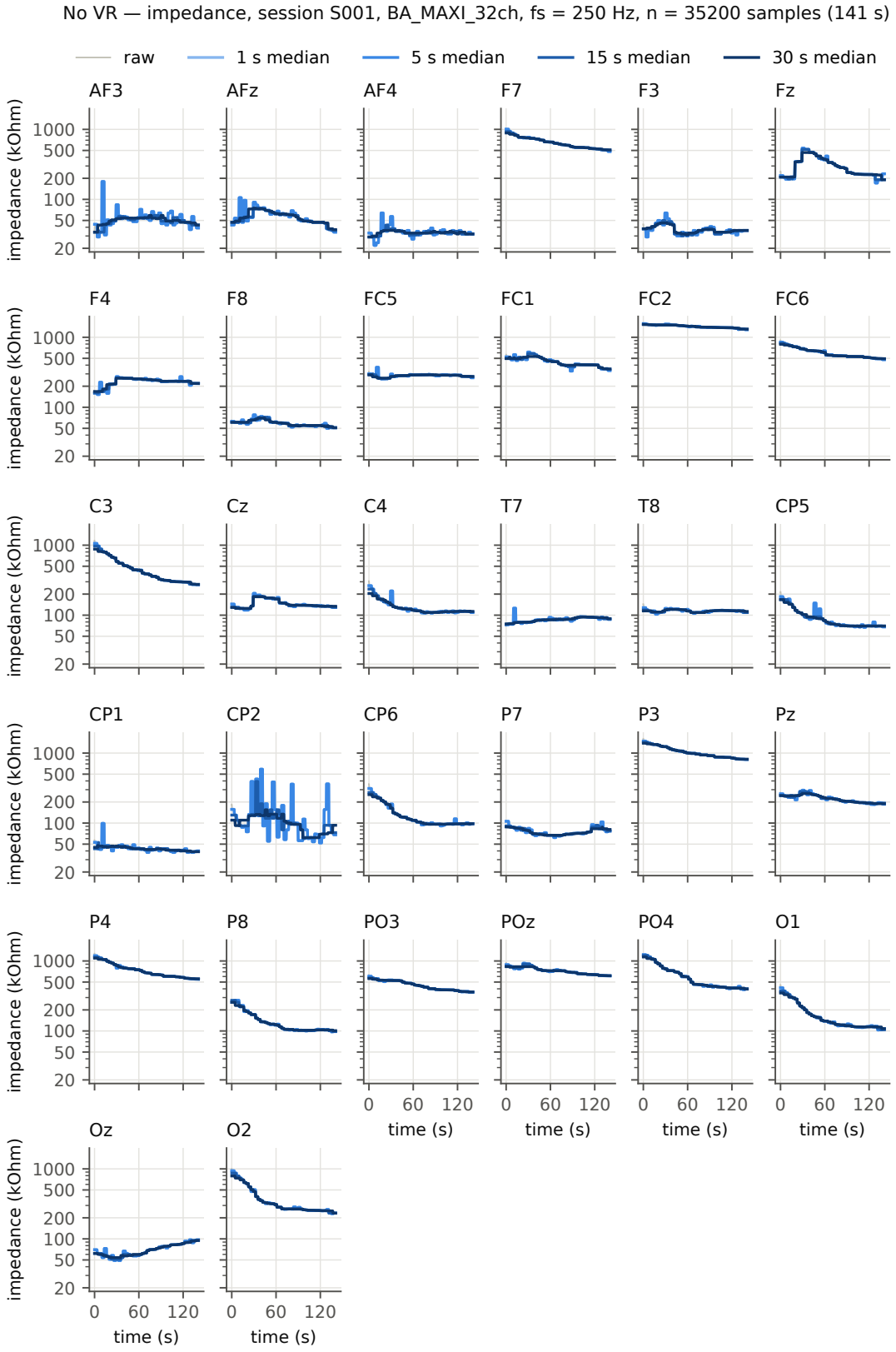


Figure 11: MAXI, no VR; as Fig. 4.

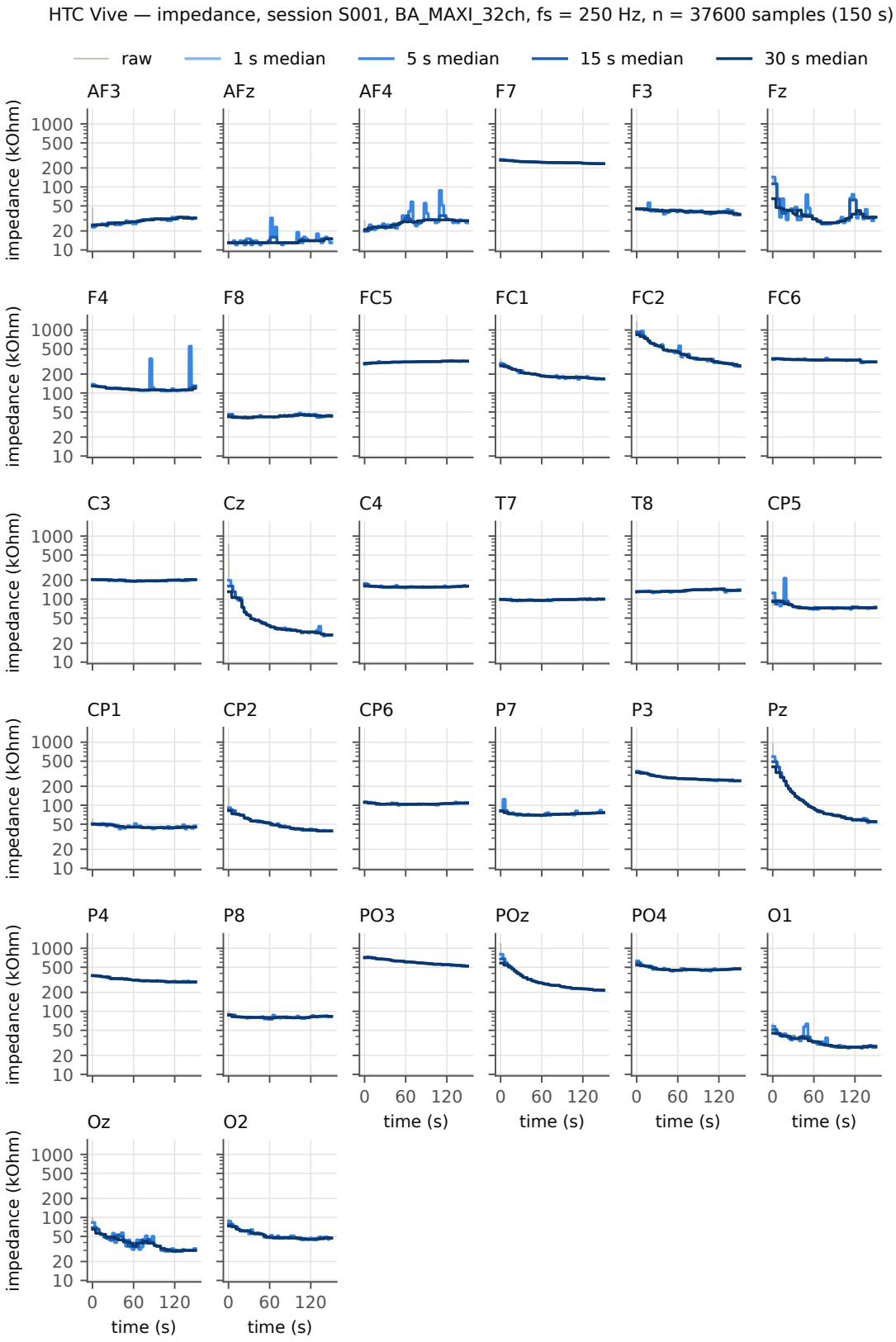


Figure 12: MAXI, HTC Vive; as Fig. 4.

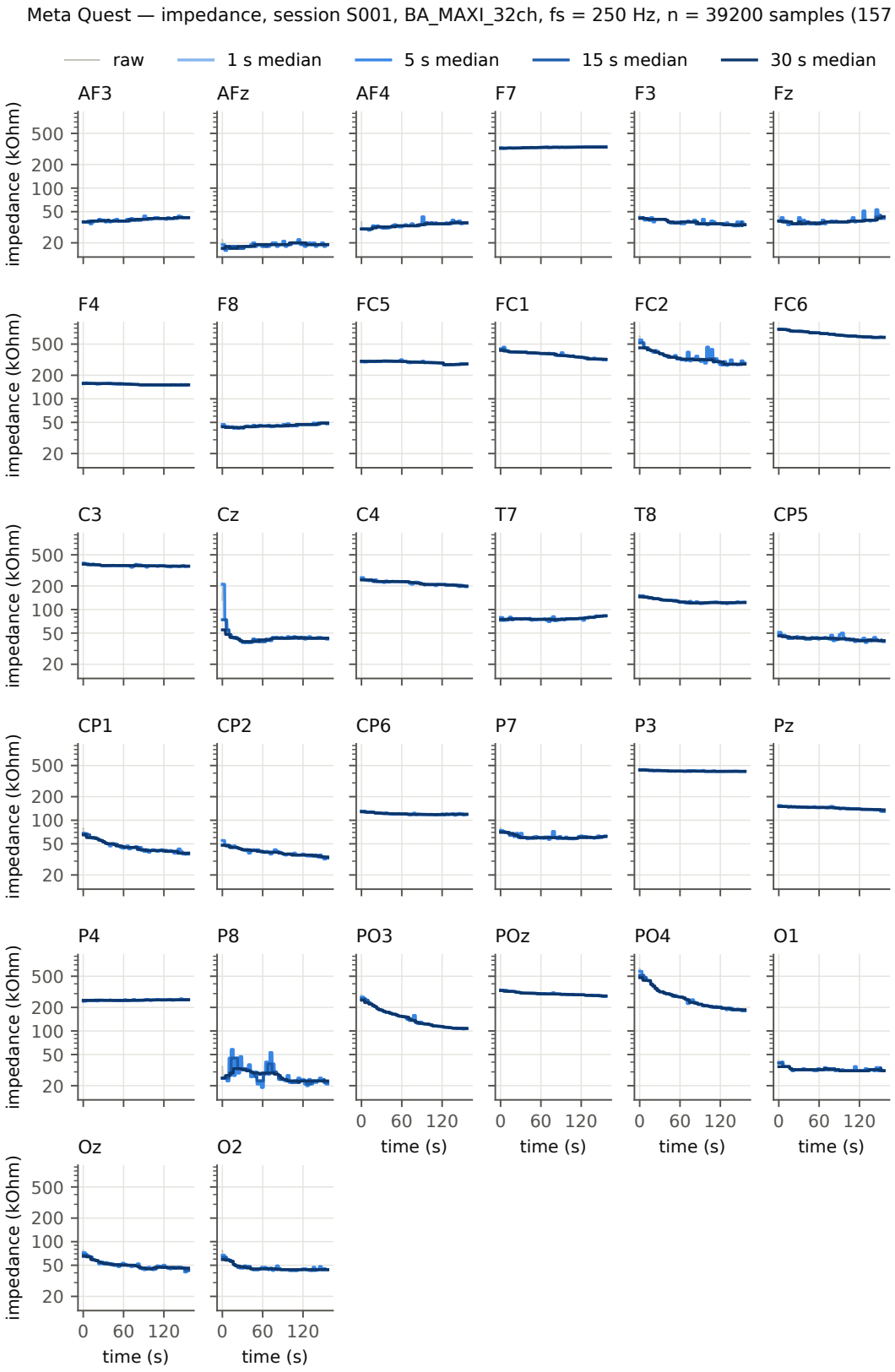


Figure 13: MAXI, Meta Quest; as Fig. 4.

B Data and reproducibility

Generated by `processing/impedance/impedance_over_time.ipynb` from `datafiles/impedance/impedance-S001-*-BA_*.csv` (MINI, MIDI, MAXI); parameters and source checksums are in `outputs/impedance/impedance_provenance_impedance-S001-*.json`. To regenerate: `processing/impedance/run_pipeline.sh`.